OHBM 2013: Education Workshop

http://www.loni.ucla.edu/twiki/bin/view/LONI/OHBM_2013_WorkflowsEdCourse

Education Course Title: **Neuroimaging 'Big Data' Challenges and Computational Workflow Solutions**

Seattle, Washington USA, Sunday, June 16, 2013, 8:00AM – 12:00PM Pacific Time

I. Logistics

Organizer: Ivo Dinov, http://www.stat.ucla.edu/~dinov/
Place: Seattle, Washington USA
Date: Sunday, June 16, 2013.
Time: 8:00 – 12:00, with a coffee break at the following time: 10:00-10:30.
Registration: All speakers must register and pay the meeting registration fee (http://www.humanbrainmapping.org/OHBM2013/)
Agenda:

1) The Pipeline Workflow Environment, Ivo Dinov, Jack Van Horn, Pipeline Team, http://pipeline.loni.ucla.edu
2) PTSD/TBI Morphometrics using the Pipeline, David Gutman, http://www.bmi.emory.edu/davidgutman
3) Neuroimaging in Python (NiPy) Architecture, Jarrod Millman, http://nipy.sourceforge.net/nipype/
4) Single subject fMRI Workflow, Satrajit Ghosh, http://nipy.sourceforge.net
5) Pipeline system for Octave and Matlab (PSOM), Pierre Bellec, http://code.google.com/p/psom/
6) Configurable PSOM Pipeline for the Analysis of Connectomes (C-PAC), Cameron Craddock, http://Fcp-indi.github.com
7) The Swift parallel scripting language and computational neuroscience applications, Justin M Wozniak, www.ci.uchicago.edu/swift
8) Conclusion/Evaluations (http://pipeline.loni.ucla.edu/learn/training/evaluation-ohbm/)

II. Pipeline/Workflow Architectures – synergistic features and complementary differences

Workflow Management System	Asynchronous Task Management	Requires Tool Recompiling	Data Storage	Platform Independent	Client-Server Model	Grid Enabled	Application Area	Dynamic Job Status Monitor	Open Source License Y/N	Scriptable language	Graphical Interface
LONI Pipeline [**http://pipeline.loni.ucla.edu**](http://pipeline.loni.ucla.edu)	Y	N	External	Y	Y	Y (DRMAA API)	Neuroscience Informatics Genomics	Y	Y plugins N core	Y bash	Y
Neuroimaging in Python Pipelines and Interfaces (Nipype) [**http://nipy.org/nipype**](http://nipy.org/nipype)	Y	N	Shared location	N (Unix, Linux)	N	Y	Neuroscience informatics	Y	Y/BSD	Y (Python)	N
Pipeline System for Octave and Matlab (PSOM) [**http://code.google.com/p/psom**](http://code.google.com/p/psom)	Y	N	Shared location	Y (Linux, Mac, Windows)	N	Y	Neuroscience informatics	Y	Y/MIT	Y (Matlab Octave)	N
SWIFT [**http://www.ci.uchicago.edu/swift**](http://www.ci.uchicago.edu/swift)	Y	N	Automated transport	Y (Linux, Mac, Windows)	Y	Y	Generic Scripting	Y	Y	Y	N
Configurable pipeline for the analysis of connectomes (C-PAC) [**http://fcp-indi.github.com**](http://fcp-indi.github.com)	Y	N	External	Y	N	Y (SGE, PBS, SLURM)	Neuroscience informatics	Y	Y/MIT	Y (Python)	Y

III. Examples of Pipeline workflows to be demoed

Workflow Management System	Pipelines	Description
LONI Pipeline **http://pipeline.loni.ucla.edu**	Global Shape Analysis Workflow (http://ucla.in/zRA2H0)	Takes raw un-skull-stripped volumes and generates 56 ROI parcellations and a scene file of group differences.
	Local Shape Analysis (LSA) Workflow (http://ucla.in/xldl85)	Takes raw volumes in 2 groups/populations and an ROI index. Generates a mean ROI shape and a color-coded maps of local shape-measure (e.g., displacement, atrophy, Jacobian, curvature, etc.)
	Genomics Computing: Mapping and Assembly, (MAQ), SAMtools, Bowtie, CNVer Workflow (http://ucla.in/yPnrk7)	The workflow contains the first step of a genomics data analysis protocol processing Illumina sequence data.
Neuroimaging in Python Pipelines and Interfaces (Nipype) **http://nipy.org/nipype**	Resting Workflow	Takes resting state time series and preprocesses the data, and computes region time courses for connectivity analyses.
	Diffusion Workflow	Takes DICOM input, performs quality control and necessary processing for tractography.
	Quality Measures Workflow	Takes resting, diffusion and structural data, computes quality measures, inserts into a queriable database and returns a PDF.
Pipeline System for Octave and Matlab (PSOM) **http://code.google.com/p/psom**	fMRI preprocessing (http://www.nitrc.org/plugins/mwiki/index.php/niak:FmriPreprocessing)	Takes raw fMRI and T1 datasets, and preprocess them, including T1/stereotaxic and T1/BOLD coregistration as well as several strategies of correction of spatially structured noise.
	Region growing (http://www.nitrc.org/plugins/mwiki/index.php/niak:RegionGrowing)	Takes preprocessed fMRI datasets of several subjects and runs a region growing algorithm (spatially constrained hierarchical clustering) to generate homogeneous regions of a fixed size (Bellec et al. 2006)
	Graph properties (http://www.nitrc.org/plugins/mwiki/index.php/niak:Connectome)	Takes a brain parcellation and preprocessed fMRI datasets, and generates connectomes as well as several graph properties (Rubinov and Sporns, 2010)
SWIFT **http://www.ci.uchicago.edu/swift**	Anatomical atlas construction	Create an averaged atlas from a collection of anatomical data.
	Spatial normalization using AIR (AIRSN)	Normalizes sets of fMRI time series of 3D volumes to a standardized coordinate system and applies motion correction and Gaussian smoothing.
	Distributed AIRSN	Illustrates the ability to distribute the processing load of a workflow across multiple distributed parallel clusters.
Configurable pipeline for the analysis of connectomes (C-PAC) **http://fcp-indi.github.com**	Anatomical and Functional Preprocessing Workflow	Takes raw functional and structural data and preprocesses them using several different strategies
	Connectome derivatives (ReHo, fALFF, centrality, voxel-mirrored homotopic connectivity, dual regression, seed correlation analysis, time series extraction)	Common functional connectivity and graph theoretic statistical derivatives for resting state data.
	Group Analysis workflow (basic anova, bootstrap analysis of stable clusters, connectome wide association studies)	Group level analysis for resting state functional connectivity and the connectome.

IV. Resource Logos

Workflow Management System	Logo
LONI Pipeline **http://pipeline.loni.ucla.edu**	
Neuroimaging in Python Pipelines and Interfaces (Nipype) **http://nipy.org/nipype**	
Pipeline System for Octave and Matlab (PSOM) **http://code.google.com/p/psom**	
SWIFT **http://www.ci.uchicago.edu/swift**	
C-PAC **http://fcp-indi.github.com**	